SRINIVASARAO PALLI

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CAREER SUMMARY

Results-driven Gen AI Architect and Senior Solution Architect with 25 years of experience in designing and delivering AI-driven, cloud-native, and enterprise-scale digital solutions. Proven expertise in Java, Spring Boot, Spring AI, and Microservices architecture, combined with hands-on experience in AWS Cloud, Generative AI, and AI Agent frameworks such as LangChain and CrewAI. Led Gen AI Organized wide Program and trainings.

Architected multiple GenAI-powered transformation initiatives, including intelligent automation platforms, AI-driven decision systems, and 5G edge computing solutions—achieving up to 30% operational cost reduction and 15% business growth through innovative cloud and AI integration.

A strategic leader skilled in end-to-end solution architecture, cloud migration, and AI/ML system design, with strong proficiency in AWS Docker/Kubernetes, and event-driven systems (Kafka, ELK). Recognized for driving cross-functional leadership, mentoring high-performance teams, and establishing AI-first architectures that enhance scalability, security, and business agility.

EDUCATION

- MCA (Regular) - 1999 Andhra University
- MTech (Distance) -2004 Amritha Viswa Vidyapeetham
- PG Program in AIML - 2021 Great Learning (Great Lakes - Texas University)
- Leader Ship Skills – ISB Hyderabad

EXPIERENCE

- Innova Solutions - Chennai 03/2024 – Present: Senior Solution Architect
- Verizon - Chennai 09/2011-03/2024: Principal Architect
- Mascon Global Limited Chennai & US 08/2000-09/2011 Software Developer, Project Lead

CERTIFICATIONS

- School of AI : Extensive Agentic AI Program 2024-25
- Crew AI : Multi Agent Framework Certification
- Educative: Certified Gen AI Architect
- Educative: Certified System Design Interview
- Educative: Certified Langchain Professional
- Educative : Certified AWS Cloud Architect

TECHNICAL SKILLS

Generative AI & Machine Learning | Cloud Architecture & DevOps | Programming & Frameworks | Databases | Orachestration | Tools

Generative AI & Machine Learning

- Gen AI, LLMs (OpenAI, Gemini, Vertex AI), Spring AI, LangChain, LangGraph, CrewAI (Multi-Agent Systems), LangSmith
- Prompt Engineering, Vector Databases (ChromaDB, Postgres Vector, Oracle 23c)
- NLP, Text Summarization, Retrieval-Augmented Generation (RAG)
- Deep Learning Frameworks: PyTorch, Keras, Scikit-Learn, NumPy

- Model Deployment & Automation: Vertex AI Pipelines, n8n Workflow Orchestration, API-based Inference, ML Flow
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Cloud Architecture & DevOps

- AWS: EC2, S3, RDS, Lambda, CloudFormation, CloudWatch, CloudTrail, ELB, DynamoDB, VPC Design
- GCP Vertex AI, Terraform, CI/CD with Jenkins and Docker
- Cloud-Native Design, Monitoring (NewRelic, ELK Stack), API Gateway (Apigee)

Programming & Frameworks

- Languages: Java, Python, JavaScript
- Frameworks: Spring Boot, Spring AI, Node.js, ReactJS, GraphQL, Selenium, Robo Framework
- Event-Driven Systems: Kafka, RabbitMQ

Databases

- Relational: Oracle, PostgreSQL, SQL Server
- NoSQL: MongoDB
- Search & Analytics: ElasticSearch
- Vector DB : Postgres Vec DB, Chroma, Pinecone,Faiss

Containerization & Orchestration

- Docker, Kubernetes, Helm (microservices deployment & scaling)

Security & Compliance

- OWASP, SOC2, Fortify, SonarQube, BlackDuck, 42Crunch
- Secure code reviews, vulnerability assessments, and compliance automation

Tools & Platforms

- RPA: Automation Anywhere
- Monitoring: NewRelic, ELK Stack
- Application Servers: WebLogic, WebSphere, Tomcat

PROFESSIONAL EXPERIENCE

SOLUTION ARCHITECT/ GEN AI SPECIALIST | Innova Solutions | Chennai, India Apr 2024 – present

Project: FinGuard AI – Generative AI-Driven Fraud Detection & Risk Scoring System

Overview:
FinGuard AI is an enterprise-grade financial fraud detection and risk intelligence system built using Generative AI and Multi-Agent architectures. It leverages LangChain, LangGraph, and LangSmith to orchestrate intelligent agents that analyze large-volume banking transactions, generate insights, and trigger automated actions across multiple data sources and APIs in real time.

Key Responsibilities

- Architected the end-to-end FinGuard ecosystem integrating LangChain (for RAG and tool orchestration), LangGraph (for agent flow and state management), and LangSmith (for evaluation, tracing, and observability).
- Designed a multi-agent pipeline where specialized agents handled data ingestion, pattern recognition, anomaly detection, and decision summarization using CrewAI and custom MCP servers.

- Integrated Oracle 23c AI with vector storage for contextual reasoning, embeddings (via Gemini Flash 2.5 and OpenAI Ada-002), and similarity search to link live transactions with past behavioral data.
- Implemented Spring Boot microservices for fraud decision APIs, exposing REST endpoints for internal FinOps dashboards and downstream applications.
- Deployed the system on AWS (EC2, RDS, Lambda, S3, CloudWatch) with containerized components using Docker and Kubernetes, enabling elastic scaling and fault tolerance.
- Integrated real-time transaction streaming via Apache Kafka, allowing ingestion from multiple financial systems and immediate scoring through AI pipelines.
- Created LangSmith monitoring dashboards to trace LLM reasoning paths, latency metrics, and accuracy scores — enhancing system transparency and audit readiness.
- Applied RAG (Retrieval-Augmented Generation) for explainable fraud summaries, enabling compliance-friendly outputs describing "why" each transaction was flagged.
- Automated Root Cause Analysis (RCA) and incident workflows by combining agent reasoning with historical RCA data and FinOps tickets.

Tech Stack

- LLM & Agentic AI: LangChain, LangGraph, LangSmith, CrewAI, Spring AI, Gemini API, OpenAI GPT-4o
- Programming & Frameworks: Python (Pydantic, LCEL), Java Spring Boot, Node.js
- Cloud & Infrastructure: AWS (Lambda, S3, CloudWatch, EC2, RDS), Docker, Kubernetes
- Database & Vector Store: Oracle 23c AI (Vector Fields), ChromaDB, Postgres
- Streaming & Analytics: Apache Kafka, ELK Stack, AWS QuickSight
- Security & Compliance: SOC2, OWASP, SonarQube, Fortify

Project: LLM-Driven Synthetic Data Generator (TableGroup)

Role: Solution Architect | Tech Stack: Python, LangChain, LLMs (Gemini), SDV, Oracle 23c AI, ChromaDB,, ReactJS

Overview

The TableGroup Synthetic Data Generator is an AI-driven data generation and augmentation platform designed to produce realistic, privacy-preserving datasets for analytics and model training.
It combines the open-source SDV framework (for statistical synthesis) with LLM-powered schema understanding and contextual data expansion, allowing enterprises to generate complete relational datasets while maintaining referential integrity and data realism.

Key Responsibilities

- Architected a hybrid synthetic data pipeline integrating SDV (for probabilistic table generation) with LLM-based contextual enrichment using LangChain and CrewAI.
- Developed a schema introspection layer that reads database metadata (from Oracle / Postgres) and constructs entity-relationship (ER) graphs to guide data dependency resolution.
- Implemented LLM-assisted prompt engineering to understand business semantics and generate domain-specific synthetic values (e.g., names, transactions, locations, behaviors).
- Utilized LangGraph to orchestrate multi-stage workflows — schema extraction, SDV modeling, LLM augmentation, validation, and data insertion.
- Integrated Oracle 23c AI vector fields and ChromaDB for similarity scoring between synthetic and real data to evaluate quality and statistical parity.
- Designed an intuitive ReactJS web UI allowing users to visualize table relationships, select target tables, configure record volumes, and trigger generation jobs.
- Used LangSmith for traceability and debugging of each synthetic generation run, ensuring consistent performance and explainability of LLM outputs.
- Enabled bulk data push to databases (Oracle / PostgreSQL / MySQL) through parallel Python data writers with automatic primary-foreign key resolution.

Project: Advanced Analytics to Improve Inbound Logistics Operations – Volkswagen

Role: Principal Architect (AI/ML & GenAI) | Tech Stack: Python, LangChain, Vertex AI, PyTorch, Oracle 23c AI, Pandas, AWS Lambda

Overview

Designed and implemented an AI-powered predictive analytics and risk classification system for Volkswagen's inbound logistics operations, enabling proactive risk mitigation and decision support across global supply chains.
The solution combined traditional ML pipelines with Generative AI reasoning to predict supply chain disruptions, assess risk levels, and provide actionable recommendations through natural-language dashboards.

Key Responsibilities

- Architected a hybrid AI system combining predictive models (built with PyTorch and scikit-learn) and Generative AI reasoning using LangChain and Vertex AI for contextual decision support.
- Designed the data ingestion and transformation pipeline using AWS Lambda, S3, and Oracle 23c AI, integrating structured (ERP) and unstructured (shipment documents, emails) data.
- Implemented RAG (Retrieval-Augmented Generation) pipelines to contextualize model outputs with logistics documents, supplier profiles, and historical delay patterns.
- Built risk scoring models using time-series and ensemble algorithms (RandomForest) to predict lead-time deviations and route bottlenecks.
- Integrated LangGraph-based reasoning layer that allowed agents to explain predictions, summarize key risks, and auto-generate incident summaries for management.
- Utilized LangSmith for model traceability, monitoring LLM responses and accuracy during validation cycles.
- Created sample Excel datasets (synthetic + real) for model validation and demo automation.

Internship Leadership: GenAI & Deep Learning Cohort – Innova Solutions

Role: Mentor & Program Architect | Teams: 6 | Interns: 28 | Theme: *"Gen AI/DL in Action!"*

Led and mentored a cohort of 28 interns across six innovation teams, driving Generative AI and Deep Learning projects from ideation to successful delivery. Designed and implemented a structured internship program that emphasized hands-on problem-solving, LLM orchestration, and enterprise AI system design.

Key Responsibilities & Achievements

- Architected six end-to-end AI solutions using LangChain, LangGraph, CrewAI, Spring AI, PyTorch, and SDV frameworks.
- Mentored interns on LLM prompt engineering, RAG pipeline development, model fine-tuning, CI/CD automation, and observability.
- Established a GenAI DevOps environment integrating LangSmith tracing, version control, and workflow observability.
- Conducted weekly design reviews, innovation challenges, and demo showcases to ensure continuous learning and measurable impact.
- Partnered with senior leadership to align projects with Innova's GenAI Center of Excellence roadmap for enterprise adoption.

🚀 Projects Delivered

1. AYS Ticket Classifier – AI-Powered Service Desk Automation
   Designed a GenAI-based ticket classification system to auto-categorize IT issues, extract key insights, and recommend routing using contextual understanding.
   Tech Stack: LangChain, LangGraph, OpenAI GPT-4o, Spring AI, Python (FastAPI), Oracle 23c AI (Vector Search), Kafka, ChromaDB, LangSmith
2. Automated Test Case Generator
   Built an intelligent code analyzer that parses repositories, reads user stories, and auto-generates test cases and validation steps using LLM reasoning.
   Tech Stack: LangChain, CrewAI, Python, Selenium, PyTest, LangSmith, GitHub Actions, OpenAI API, SDV for synthetic test data

3. CI/CD Pipeline Analyzer
   Developed an LLM-driven DevOps assistant that reviews Jenkins/GitHub pipelines, detects vulnerabilities, and suggests configuration optimizations.
   Tech Stack: Spring Boot, LangChain, Spring AI, CrewAI, Jenkins API, GitHub REST API, Python, Docker, AWS Lambda, Grafana
4. Observability Analyzer
   Created an AI agent that consolidates metrics/logs from ELK stack and CloudWatch, identifies anomalies, and explains potential root causes through natural language insights.
   Tech Stack: LangChain, OpenAI GPT-4o, ELK Stack, AWS CloudWatch, Python, FastAPI, LangSmith, ChromaDB, Grafana
5. News Feed Generator
   Built a Generative AI pipeline for creating personalized, category-based news summaries with sentiment control and tone optimization.
   Tech Stack: LangChain, Gemini API, OpenAI GPT-4o, Python, Streamlit, MongoDB, Pinecone, Hugging Face Transformers
6. Attendance System using Facial Recognition
   Implemented a computer-vision solution for real-time attendance tracking with face detection, recognition, and emotion analytics.
   Tech Stack: PyTorch, OpenCV, Flask, FaceNet, SQLite, LangChain (for description generation), AWS S3 (for image storage), Streamlit UI

PRINCIPAL ARCHITECT | Verizon | Chennai Sep 2011 – Mar 2024

Project: Wireline GCH2.0 Migration Application (Team Size: 40)
Technologies Used: SpringBoot, Tomcat, AWS (EC2, S3, RDS), Docker, Kubernetes, ElasticSearch, RabbitMQ, Jenkins, Terraform

Project Overview: Designed and led the end-to-end migration of Verizon's Wireline GCH (Global Customer Hierarchy) system from GCH1.0 to GCH2.0. The project aimed to modernize and improve the scalability of Verizon's infrastructure by migrating to a microservices architecture, enabling faster customer data retrieval, and providing a 360-degree view of customer accounts and revenue information.

Key Responsibilities

- End-to-End Solution Design: Architected a microservices-based solution leveraging Spring Boot and AWS for scalability and resilience. Implemented best practices in distributed systems, ensuring high availability and fault tolerance.

- Data Pipeline Optimization: Built a high-throughput data pipeline using ElasticSearch and AWS services to handle high volumes of customer and revenue data. Developed a synchronization framework to keep databases and ElasticSearch in sync within seconds.

- Performance Tuning: Conducted extensive load testing and query optimization, which reduced data retrieval time by over 50%. Fine-tuned ElasticSearch indices and optimized RabbitMQ configurations to handle peak traffic with zero downtime.

- Automation and CI/CD: Automated deployment and testing using Jenkins and Docker, reducing deployment times by 40% and improving overall team efficiency. Employed Kubernetes for container orchestration, allowing for flexible scaling based on demand.

- Stakeholder Communication: Worked closely with Verizon's FP&A (Financial Planning & Analysis) team to align migration objectives with business goals, ensuring accurate reporting of customer revenue data at both account and
- master account levels.

- Improved Data Access: Achieved a 50% reduction in data retrieval time and improved system responsiveness, significantly enhancing user experience.

- Enhanced Data Consistency: Implemented a near-real-time synchronization mechanism between ElasticSearch and databases, maintaining a maximum lag of 20 seconds, which greatly improved data consistency.

- Cost Efficiency: Leveraged AWS auto-scaling and optimized resource usage, resulting in an infrastructure cost savings of approximately 30%.

- Operational Resilience: Achieved zero downtime during the migration process, ensuring continuous service availability for Verizon's critical customer hierarchy operations.

PROJECT LEAD | MGL Chennai & US Jun 2000 – Aug 2011

Roles performed as Software developer, Lead and Project Manager

- Developed and Led Projects during this period.
- Onsite Experience: Worked onsite in the USA for over 5 years, providing direct support and collaboration for Walgreens, a key client